MILITARY SPECIFICATION

PLOTTING BOARD, INDIRECT FIRE: M16 W/E

1. SCOPE

1.1 This specification covers Indirect Fire Plotting Board M16 with equipment consisting of Plotting Board Case M105 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL–F–13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL–P–14232 —Parts, Equipment and Tools for Ordnance Materiel, Packaging of.

MIL–I–45208 —Inspection System Requirements.

STANDARDS

Military

MIL–STD–105 —Sampling Procedure and Tables for Inspection by Attributes.

MIL–STD–109 —Quality Assurance Terms and Definitions.

DRAWINGS

U.S. Army Munitions Command

F7633931—Plotting Board, Indirect Fire: M16.

F7680317—Case, Plotting Board: M105.

PACKAGING DATA SHEET

MIL–P–14232/ P8270330 —Packaging of Plotting Board, Indirect Fire: M16 W/E.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Materials.** Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.2 **Fabrication.** The plotting board shall be manufactured in accordance with Drawing F7633931 and drawings pertaining thereto. The plotting board case shall be manufactured in accordance with Drawing F7680317 and drawings pertaining thereto.

3.3 **General specification.** The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL–F–13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Part identification and marking.
(e) Workmanship.

3.4 **Environmental.**

3.4.1 *Storage temperatures.* The plotting board shall show no evidence of physical failure and shall meet all performance require-

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ments of this specification when thermally stabilized at a standard ambient temperature of between 60° and 90° Fahrenheit (F) after having been exposed to, and thermally stabilized at, ambient temperatures of −60° and +140° F.

3.4.2 *Operating temperatures.* The plotting board shall operate freely while exposed to, and thermally stabilized at, ambient temperatures of −40° and +125° F, subsequent to which the plotting board shall meet the requirements of 3.5 to 3.9 inclusive when checked at a standard ambient temperature of between 60° and 90° F.

3.5 **Accuracy.** With the conditions existing as designated in (a) through (c) below, and with checkpoints located on the grid 2050 meters to the right and left of the "0" disk line and on the 2050 meter line in each quadrant (total of four points) the board shall meet the requirements specified in 3.5.1 through 3.5.4.

(a) Pivot of the disk engaged by the detent.

(b) Pivot button of the deflection arm inserted in the socket of the disk pivot.

(c) The "0" azimuth index line on the disk in coincidence with the "0" index line on the base grid.

3.5.1 With the disk rotated 800 mils, the index lines on the disk shall intersect the four located points on the base grid within 5 mils.

3.5.2 The deflection arm scale, when rotated to the four established points, shall indicate 2,900 meters ±10 meters.

3.5.3 The deflection arm meter scale when superimposed on the base grid meter scales shall coincide at the 3100 meter lines within 5 meters.

3.5.4 The vernier scales of the base grid and arm shall coincide within 0.5 mil.

3.6 **Rigidity.** Subsequent to applying a 40-pound load equally distributed on opposing sides (20 pounds per side) at the outer edge of the board for a period of 5 minutes, with the board center supported by a 3-inch diameter tube, and with the board thermally stabilized at a standard ambient temperature of between 60° and 90° F, there shall be no permanent deformation of the plotting board.

3.7 **Warpage.**

3.7.1 The maximum warp between the disk and the base of the plotting board shall not exceed 0.062 inch as measured at the outer edge of the disk.

3.7.2 The maximum warpage of the deflection arm of the plotting board shall not exceed 0.03 inch, and this warpage shall not extend in from the edges by more than 3/16 inch.

3.8 **Surface finish.** The top surfaces of the disk and the base grid of the plotting board shall allow clear pencil markings and clean erasures as specified on pertinent drawings. The materials shall be translucent and have no defects such as pits, digs, scratches with depth, stains, or waves on the surface that may be mistaken for pencil marks. The lines and figures on the base grid, as seen through the disk, shall be clearly legible.

3.9 **Operability.** With the pivot of the disk engaged by the detent, and with the pivot button of the deflection arm inserted in the socket of the disk pivot, the arm and disk shall be capable of rotating freely. At a standard ambient temperature of between 60° and 90° F the arm and the disk shall be removable from the base with normal finger pressure and without causing permanent distortion.

3.10 **Plotting board case M105.** The case shall conform to all requirements specified on Drawing F7680317 and drawings pertaining thereto.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his

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own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *General provisions.* The quality assurance provisions of this specification and of other documents referenced herein form the basis for inspection to be performed by the supplier. Inspection shall be in accordance with the requirements of Specification MIL–I–45208 and the component inspection requirements of Specification MIL–F–13926. Definitions of inspection terms not otherwise defined herein shall be as listed in Standard MIL–STD–109.

4.2 Inspection provisions.

4.2.1 *Inspection lot.* Unless otherwise specified by the contracting officer, inspection lot size, formation, and presentation of lots shall be in accordance with "Submission of Product" as specified in Standard MIL–STD–105.

4.2.2 *Examination and tests.* Components and subassemblies shall be inspected in accordance with the inspection provisions contained in Specification MIL–F–13926. Examinations and tests related to Section 3 herein shall be performed on a defect (individual characteristic) basis in accordance with Standard MIL–STD–105, and the inspection level and sampling plans specified in Table I titled "Classification of Defects." Examinations and tests for Packaging, Packing and Marking shall be in accordance with Specification MIL–P–14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

TABLE I. *Classification of defects*

Use Inspection Level II of Table I with
Sampling Plan Table II–A of Standard MIL–STD–105

		Requirement	*Test procedure*
Critical: None.			
Major:			
	AQL 1.0 percent defective		
101.	Accuracy	3.5	4.5.3
102.	Rigidity	3.6	4.5.4
103.	Warpage (disk)	3.7.1	4.5.5
104.	Warpage (arm)	3.7.2	4.5.6
105.	Surface finish	3.8	4.5.7
106.	Operability	3.9	4.5.8
107.	Carrying case	3.10	4.5.9
Minor:			
	AQL 4.0 percent defective		
201.	Open spots in filled graduations	3.3(e)	Visual

4.2.3 *Disposition of non-conforming product.* Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in Standard MIL–STD–105.

4.3 Control tests. Three plotting boards shall be selected at random from each 100 produced, or from each month's production, whichever occurs first. Control samples shall have successfully met all acceptance tests specified herein prior to conducting the following examinations and tests:

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Control test	*Requirement*	*Test procedure*
Materials	3.1	Drawings—Material Specifications
Fabrication	3.2	4.5.1
General specification	3.3	Visual, MIL–F–13926
Storage temperatures	3.4.1	4.5.2
Operating temperatures	3.4.2	4.5.2

4.3.1 *Control test failure.* Should the control test sample fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved.

4.4 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall furnish and maintain all required measuring and testing equipment in accordance with the "Measuring and Test Equipment" requirement of Specification MIL–I–45208. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 *Government furnished inspection equipment.* Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with "Measuring and Test Equipment" and "Government Furnished Material" requirements of Specification MIL–I–45208.

4.4.2 *Contractor furnished inspection equipment.*

4.4.2.1 *Contractor design.* The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5, and with the "Component Inspection" requirements of Specification MIL–F–13926. Concept, construction, materials, dimensions, and tolerances used in design of test equipment shall be so selected and controlled as to insure that the test equipment will permit positive rejection of a product which exceeds the prescribed tolerance limit, and will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit. Construction shall be such as to facilitate routine calibration of the equipment.

4.4.2.2 *Devices used for environmental testing.* All environmental testing devices used to determine conformance with the temperature requirements of 3.4 shall conform to the "Test Facilities" requirements outlined in Specification MIL–F–13926.

4.5 Test methods and procedures.

4.5.1 *Fabrication.* The plotting board and case shall be subjected to a visual, tactile, and dimensional inspection for conformance to the applicable drawings and requirements of this specification. Dimensional inspection shall be performed using standard measuring equipment.

4.5.2 *Storage and operating temperatures.* This test shall be conducted in accordance with Procedure I of Specification MIL–F–13926 except that the temperatures of 3.4.1 and 3.4.2 shall apply. While stabilized at each temperature extreme specified in 3.4.2, the plotting board shall be operable, without binding, when normal finger pressure is applied to move the arm and disk around the pivot. Upon completion of the specified exposures and return to standard ambient temperature (60° to 90° F), the plotting board shall be visually inspected for damage. The board shall then be subjected to the tests specified in 4.5.3 through 4.5.8 and shall conform to the requirements specified therein.

4.5.3 *Accuracy.* The accuracy tests are visual tests performed with the aid of a 10X magnifier in accordance with the conditions indicated in 3.5. The accuracy requirements shall be within the tolerances specified in 3.5.

4.5.4 *Rigidity.* The rigidity test shall be conducted in accordance with the conditions indicated in 3.6. The direct load of 20 pounds on each side shall be applied simultaneously, diametrically opposite each other on the outer edge of the board. This test shall be repeated

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three times to determine conformance with the requirements of 3.6. Upon completion of this test the board shall be inspected for permanent deformation in accordance with the test specified in 4.5.5.

4.5.5 *Warpage (disk).* Warpage between the disk and the base shall be checked by measuring the distance between the disk and the base, at the perimeter of the disk, by means of standard measuring equipment. The highest measurement shall be considered maximum warp and shall not exceed the tolerance specified in 3.7.1.

4.5.6 *Warpage (arm).* Warpage of the deflection arm shall be checked by measurement of the top surface of the arm with standard measuring equipment after the arm has been located on a flat surface plate. The difference between the highest and lowest measurement shall be considered warp and shall not exceed the limits specified in 3.7.2.

4.5.7 *Surface finish.* The top surfaces of the disk and base of the plotting board shall be capable of accepting clear and legible markings from a lead pencil of a maximum hardness of 2H under normal writing pressure. Upon erasure with a soft commercial eraser, there shall be no evidence of permanent marks, smudges, or smears. The plastic surfaces of the disk and base shall be visually inspected for defects, opaqueness, and legibility of lines and figures in compliance with the requirements specified in 3.8.

4.5.8 *Operability.* The test consists of a tactile examination for rotation of the arm and the disk around the pivot point, and for removal of the arm and disk from the base with normal finger pressure. In compliance with the requirements specified in 3.9, there shall be no binding when the arm and the disk are completely rotated around the pivot, and removal of the arm and disk from the pivot shall cause no permanent distortion.

4.5.9 *Plotting board case M105.* The case shall be subjected to a visual, tactile, and dimensional inspection to determine conformance to 3.10 and applicable drawings.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with Specification MIL–P–14232 and Packaging Data Sheet MIL–P–14232/P8270330. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. Plotting Board M16 is used to plot data for indirect fire of mortars.

6.2 Ordering data. Procuring documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of an applicable level of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).

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